PRODUCT REQUIREMENTS DOCUMENT (PRD)

Han River-Incheon Water Quality

Eutrophication (Algal Bloom / Red Tide) Forecast App

Real-time monitoring and alert service

Platform: Flutter (iOS / Android)
Monetization: Fully free (no paid tier)

Document Version: v1.0

Date: June 2026

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1. Product Overview

1.1 One-line Definition

A free mobile app that traces, in real time, how nitrogen and phosphorus from Seoul's domestic wastewater travel through the Han River estuary and trigger eutrophication (algal blooms / red tides) along the Incheon coast, using public environmental data to warn Incheon fishers and citizens.

1.2 Problem

- Cage aquaculture farms along the Incheon coast can suffer hundreds of millions of won in damage within days of a red tide outbreak.
- Existing government announcements are broad-area or after-the-fact, so individual farms often miss the window to evacuate or apply countermeasures (clay dispersal, oxygen supply).
- General citizens lack an intuitive way to see which coastal areas are at risk.

1.3 Solution

The system collects real-time water-quality data (water temperature, pH, total phosphorus, dissolved oxygen, etc.) from the Ministry of Environment and the National Institute of Environmental Research, compares it against biological thresholds for eutrophication, and computes a risk level. Results are visualized on a map-and-graph dashboard, and a push notification is sent when a user's registered point reaches a risk threshold.

Key change: The 'premium paid alert (SMS subscription)' feature from the original concept is fully merged into the free tier. All features, including location registration and custom alerts, are provided at no cost, and no payment or subscription module is in scope.

1.4 Goals

1. Primary: Build a working Flutter app and publish it to iOS/Android.
2. Secondary: Demonstrate public value and submit it as a school activity, competition, or academic record deliverable.
3. Tertiary: Use accumulated data to extend toward a B2G infrastructure linked with local government (Incheon Marine & Fisheries Division). Out of scope for this PRD; future work.

2. Scientific Basis (BIO Logic)

The app's risk-judgment logic transplants the eutrophication mechanism into conditional rules. The algorithm assumes the following causal chain.

2.1 Pollution Causal Chain

- **Source:** Domestic wastewater / effluent from densely populated Seoul areas contains high nitrogen (N) and phosphorus (P), entering the Incheon coast via the Han River estuary.
- **Bloom conditions:** When water temperature exceeds 25°C and phosphorus concentration rises, phytoplankton and cyanobacteria multiply exponentially, producing algal blooms / red tides.
- **Ecosystem damage:** Algae cover the surface and block sunlight; as algae decay, they deplete dissolved oxygen (DO), suffocating fish and shellfish.

2.2 Risk Thresholds (Initial Values)

The values below are initial settings and will be calibrated after real data is collected. They are kept as code constants so they can be adjusted.

Indicator	Threshold	Meaning
Water temp	≥ 25°C	Active algal growth temperature
Total phosphorus (T-P)	≥ 0.05 mg/L	Excess eutrophic nutrient
Dissolved oxygen (DO)	≤ 4 mg/L	Fish/shellfish mortality zone
pH	≥ 9.0	Indicator of excess photosynthesis (algal bloom)

Example: water temp ≥ 25°C AND T-P ≥ 0.05 → escalate risk to 'Red (high)'. A single condition met → 'Yellow (caution)'.

3. Target Users (Personas)

User	Needs	Key features
Incheon fish farmer	Know red-tide risk for my farm zone in advance and time countermeasures	Point registration, risk push alert, forecast graph
Coastal citizen / angler	See intuitively which areas are safe today	Real-time map, color-coded risk level
Teacher / judge	Verify the credibility of data sources and logic	Source attribution, judgment-rationale screen

4. Feature Requirements

Priority: P0 (MVP required), P1 (recommended), P2 (future).

ID	Feature	Description	Priority
F-01	Real-time map	Show Han estuary/Incheon coast stations on a map with risk-level color markers (green/yellow/red)	**P0**
F-02	Station detail	On tap, show water temp, pH, T-P, DO values and recent trend graph	**P0**
F-03	Risk forecast	Show current risk level and short-term forecast from BIO threshold logic	**P0**
F-04	Point registration	Users register farm/area of interest (free, no count limit)	**P0**
F-05	Risk push alert	Push when a registered point hits a risk threshold (former paid feature, now free)	**P0**
F-06	Judgment rationale	Screen showing current vs threshold values explaining why it is risky	**P1**
F-07	Source attribution	Show data source (Ministry of Environment / data portal) and update time	**P1**
F-08	History view	Timeline of past water quality / red-tide events per point	**P2**
F-09	Gov. integration	B2G data-sharing expansion (future)	**P2**

4.1 Non-functional Requirements

- Update cadence: follow the public-data refresh cycle, with at least hourly in-app sync.
- Offline: cache the last sync so data can be viewed without a network connection.
- Performance: map initial load within 3 seconds (cached).
- Privacy: location registration stores coordinates only, minimizing personally identifiable data.

5. Technical Architecture (CS)

5.1 Data Flow

[Public Data API] → [Backend collect / process / judge] → [Supabase storage] → [Flutter app query / push]

4. Collect: a backend scheduler periodically calls the public-data portal water-quality API.
5. Process: handle missing values and normalize, then compute risk level via BIO threshold logic.
6. Store: load stations, water-quality time series, and risk levels into Supabase (PostgreSQL).
7. Deliver: the Flutter app queries Supabase and sends a push when risk changes.

5.2 Tech Stack

Layer	Technology	Notes
Mobile app	Flutter (Dart)	Single codebase for iOS / Android
Map	google_maps_flutter, etc.	Station markers, risk colors
Backend / collect	Python + scheduler	API crawl, Pandas processing, judgment
DB / Auth	Supabase (PostgreSQL)	Water-quality series, registered points
Push	Firebase Cloud Messaging	SMS replacement, free
Data source	Public Data Portal water API	Ministry of Environment / NIER

Note: if a laptop-only demo is needed, the same logic can also be built as a Python/Streamlit web dashboard, but the deliverable of this PRD is the Flutter app.

5.3 Data Model (Summary)

Table	Key columns
stations	id, name, lat, lng, region
measurements	station_id, ts, water_temp, ph, total_p, dissolved_o2
risk_status	station_id, ts, level(green/yellow/red), reason
user_locations	user_id, label, lat, lng, created_at
alerts	user_id, location_id, level, sent_at

6. Screens

Screen	Content
1. Map home	Han River-Incheon coast map, station markers, risk colors, refresh time
2. Station detail	Water temp / pH / T-P / DO values, recent trend graph, risk badge
3. Risk explanation	Current vs threshold comparison, judgment rationale text
4. Registered points	List of saved points, add/delete, current level per point
5. Alert history	Timeline of sent risk alerts
6. Info / sources	Data sources, logic explanation, disclaimer

7. Development Roadmap

Phase	Target duration	Deliverables
Phase 1	1-2 weeks	Public data API integration, collection script, Supabase schema
Phase 2	2-3 weeks	BIO judgment logic, risk-level computation pipeline
Phase 3	3-4 weeks	Flutter map / detail screens (F-01 to F-03) MVP
Phase 4	2 weeks	Registered points / push alerts (F-04 to F-05), FCM integration
Phase 5	1-2 weeks	QA, store deployment, source/disclaimer screen, submission packaging

7.1 MVP Scope (Minimum for Launch)

- F-01 real-time map + F-02 detail + F-03 risk forecast + F-04 registered points + F-05 push.
- Limit stations to core Han estuary-Incheon coast stations to start.

7.2 Risks

- Infrequent updates or frequent gaps in the public API may lower forecast accuracy; mitigate with caching/interpolation.
- Thresholds may not match the actual sea area; calibration is needed after accumulating real data.
- False positives/negatives in push alerts erode trust; clearly label outputs as 'forecasts' early on and add a disclaimer.

7.3 Success Metrics (MVP)

- App store registration completed and a normal build runs.
- Core station data refreshes and displays periodically.
- After point registration, a push is actually sent when a threshold is reached.